Kanav Chopra

AI Researcher | Full-Stack Developer | User-Centric Applications

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SUMMARY

Software Engineer with 6 years of experience building scalable full-stack web applications. Experienced in React, TypeScript, Next.js, Node.js, and Python, with hands-on work on LLM-powered healthcare and enterprise systems, legacy modernization, and performance optimization.

TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Vite.js, Vue.js

Backend: Python, FastAPI, Node.js, PostgreSQL, SQL, Docker, Kubernetes, GCP, AWS

Databases: PostgreSQL, SQLite, Vector Database, MongoDB, Supabase

AI / ML: LLM integration, Prompt Engineering, Generative AI, Agentic AI, Machine Learning, TensorFlow, Vector Embeddings, ReAct Patterns, Tool Orchestration, Evaluation Pipelines

Tools & Infra: Git, GitLab, CI/CD, Webpack, GraphQL, REST APIs, Server-Sent Events (SSE), WebSockets

EXPERIENCE

AI Researcher / Software Engineer — Stanford University 06/2025 – Present

Palo Alto, CA

- Led full-stack development and launched **SAGE** (sage.arise-ai.org), a clinician decision-support tool, reducing consulting time by 70–80% (from 15 min to 2–3 min).
- Engineered prompt versioning, feedback capture, and session tracking in React, TypeScript, Node.js, and Python, accelerating prompt experimentation cycles by 40% for AI-assisted clinical notes.
- Designed backend services to log 10K+ user interactions, manage model prompt configurations, and power feedback-driven iteration that improved response quality scores by 25%.
- Architected and launched **Stanford Health Path**, an LLM-powered patient navigation tool, increasing patient engagement with recommended routes by 55%.
- Launched the ARISE AI research network website using Next.js and a Headless WordPress CMS (arise-ai.org).

Senior Software Engineer — Vensure Employer Solutions 03/2023 – 08/2024

Noida, UP, India

- Led modernization of legacy systems by transitioning to a microservices architecture, reducing technical debt by 40%.
- Selected and implemented modern technologies, improving system efficiency by 50%.
- Architected and deployed robust HR Management system features using Next.js, Node.js and TypeScript.
- Designed visually appealing, intuitive, and user-friendly interfaces for web and mobile applications using React.js.
- Optimized UI performance by 60% using minification, lazy loading, and reduced network requests.
- Planned a scalable full-stack architecture from scratch aligned with product requirements.
- Reviewed and refactored large legacy codebases and mentored junior developers through code reviews, resolving critical issues and significantly reducing application crashes.

Software Developer — Khojdeal 08/2019 – 03/2023

Gurugram, Haryana, India

- Developed multiple content-based websites using CMS system with complete Core Web Vitals optimization up to 100%.
- Partnered with the marketing team on Google Tag Manager (GTM) and Google Analytics (GA) integration, improving conversion rate by 25%.
- Deployed multiple websites through GitLab CI/CD pipelines.
- Architected and deployed robust HR Management system using Python, Typescript and Next.js.
- Integrated Cloudflare across organizational and client websites to improve security and DNS management.
- Built complex, fully functional e-learning simulations using pure JavaScript and Typescript.

- Developed applications consuming JSON and XML for RESTful web services using JavaScript.
- Converted legacy Flash applications to HTML5 and JavaScript, compatible across mobile devices and browsers, resolving major browser compatibility issues.

PROJECTS

SAGE (sage.arise-ai.org)
A secure, full-stack medical consultation and referral system using real-time, multi-model AI orchestration to deliver immediate, contextual clinical guidance.

- Architected a Multi-Model AI Orchestration System (FastAPI), integrating Claude, Gemini, and Azure OpenAI for dynamic, low-latency switching to optimize clinical accuracy and cost-efficiency.
- Implemented a Server-Sent Events (SSE) pipeline with React/TypeScript and leveraged vector embeddings for high-accuracy clinical context matching, significantly reducing LLM hallucination.
- Integrated PostgreSQL for session management with strict CORS and Pydantic validators for secure handling of sensitive clinical data.
- Built a modern, multi-step React/TypeScript/MUI interface with stringent input validation to enforce data quality for clinicians.

Clinical Research Multi-Agent System | Python, LangGraph, Vertex AI (https://github.com/code-kanav/clinical-research-agent)

- Built a 5-agent LangGraph pipeline (planner, search, reader, critic, synthesizer) implementing ReAct, self-reflection, and hierarchical delegation across PubMed, Semantic Scholar, and arXiv.
- Swappable Claude/Gemini-on-Vertex providers via ADC; eval harness scoring citation accuracy, faithfulness, latency, and per-query token cost.

PUBLICATIONS

- 4 peer-reviewed publications (AMIA 2025, PSB / Biocomputing 2026) on LLM evaluation and AI-assisted clinical decision support — Google Scholar.

EDUCATION

Master of Science in Cybersecurity and Artificial Intelligence August 2024 – May 2026
Webster University — St. Louis, Missouri, USA

Bachelor of Science in Computer Science August 2015 – May 2019
University of Petroleum and Energy Studies(UPES) — Dehradun, UK, India